

A Learning-Based Cooperative Navigation Approach for Multi-UAV Systems Under Communication Coverage

Di Wu, *Member, IEEE*, Zhuang Cao , Xudong Lin , Feng Shu , *Member, IEEE*, and Zikai Feng

Abstract—In multi-unmanned aerial vehicle (UAV) systems cooperative navigation under the communication coverage of ground base stations, UAVs encounter challenges in maintaining reliable communication, ensuring flight safety, and achieving efficient collaboration. To address these challenges, this study formulates the cooperative navigation problem as a Markov game and introduces a two-stream graph multi-agent proximal policy optimization (two-stream GMAPPO) algorithm based on the graph neural network (GNN). We model UAVs and other entities as graph-structured data, aggregate the node information of these graph-structured data through the GNN module, and extract potential features related to UAVs. This allows UAVs to obtain richer local information. Through the two-stream network structure, the extracted potential features related to UAVs are combined with the state space of UAVs to enhance adaptability to environmental changes and improve navigation safety. Simulation results demonstrate that the proposed method significantly outperforms baseline algorithms in both mean reward and convergence speed, confirming the superior performance of two-stream GMAPPO.

Index Terms—Multi-UAV, navigation, communication coverage, graph neural network, reinforcement learning.

I. INTRODUCTION

IN RECENT years, multi-unmanned aerial vehicle (UAV) systems have become more widespread and influential in society due to their flexibility and ease of deployment [1], [2].

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Multi-UAV systems are widely applied in various fields, such as agriculture [3], environmental monitoring [4], and disaster relief [5]. Furthermore, with the continuous advancement of communication and UAV technologies, multi-UAV systems also play an irreplaceable role in communication networks [6], [7]. Ground base stations (BSs) often suffer significant damage in emergencies, leading to communication disruptions. Multi-UAV systems can function as airborne communication platforms or extend the coverage of existing ground BSs to restore connectivity in affected areas [8], [9]. Additionally, multi-UAV systems can serve as airborne equipment, exchanging information to support missions such as cargo delivery [10], remote sensing [11], and monitoring [12] by establishing communication links with ground BSs.

In multi-UAV applications, navigating UAVs to designated locations while ensuring safe and efficient flight is crucial for mission success. The enhanced ant colony optimization algorithm [13] provides a viable solution for UAV navigation in static indoor environments. However, its strong reliance on ideal conditions limits its broader applicability. The study [14] introduced a UAV navigation algorithm based on rapidly exploring random trees, determining the optimal feasible path by analyzing point cloud data from the environment. Recently, UAV systems integrating simultaneous localization and mapping (SLAM) with inertial measurement units and radar data have been widely applied in practical scenarios. Although the SLAM method enables autonomous navigation of UAVs in complex environments, it entails high computational complexity and stringent requirements for device performance. These navigation methods primarily focus on single-mission scenarios and overlook situations where UAVs must maintain communication during navigation.

Multi-UAV navigation tasks in communication scenarios impose more stringent path planning requirements. During flight, UAVs must maintain stable air-to-ground communication to ensure reliable connections with ground BSs while minimizing unnecessary movements to optimize energy efficiency. The study [15] derives the optimal and sub-optimal path solutions for UAV navigation by solving the equivalent shortest path and using a grid-based quantization method, respectively. Additionally, this method ensures that the expected signal to interference plus noise ratio (SINR) between the UAV and each associated ground BS during flight meets the target link quality, thereby achieving the UAV’s navigation task. In [16], graph theory is utilized to

solve for the shortest path between any two points that satisfy communication connectivity requirements, aiming to maintain communication connectivity and minimize mission completion time for logistics UAVs. Study [17] presents a polynomial-time approximate solution based on dynamic programming to address the navigation problem for UAVs, ensuring that disconnection from the ground BS during flight doesn't exceed a predefined time constraint. All these methods share a common feature: they require explicit path planning. Moreover, these traditional methods lack predictive capabilities, limiting their effectiveness in more complex and unknown environments.

Fortunately, due to the continuous development of machine learning, research on deep reinforcement learning (DRL) is advancing rapidly [18], offering new approaches to address the navigation missions of UAVs under the communication coverage of ground BSs. In DRL, an agent receives reward feedback by continuously interacting with the environment, with the aim of maximizing the reward and learning the optimal policy. In recent years, the actor-critic algorithm, which combines deep q-learning network (DQN) [19] and policy gradients (PG) [20], has become very popular. The actor-critic algorithm, exemplified by methods such as deep deterministic policy gradient (DDPG) [21], twin delayed deep deterministic policy gradient (TD3) [22], and proximal policy optimization (PPO) [23], allows for the simultaneous learning of value and policy functions, enhancing the efficiency and stability of the algorithms. However, in a multi-agent environment, the decisions of multiple agents affect each other, which can lead to unstable policies when DRL algorithms are applied directly. The multi-agent reinforcement learning (MARL) algorithms, such as multi-agent deep deterministic policy gradient (MADDPG) [24] and multi-agent twin delayed deep deterministic policy gradient (MATD3) [25] and multi-agent proximal policy optimization (MAPPO) [26], address the instability issues of DRL algorithms in multi-agent environments by introducing the centralized training and decentralized execution (CTDE) mechanism [24]. During training, agents have access to global information, such as the states, actions, and rewards of all agents, enabling the algorithm to learn collaborative strategies effectively and improve overall system performance. In the execution phase, however, agents rely solely on local observations to make decisions, without sharing state or policy information. Despite not having access to global information during execution, the trained agents can still make independent decisions effectively, ensuring efficient operation in the environment.

Technological advancements in DRL and MARL have also driven research into the integrating reinforcement learning with UAV navigation. In [27], a novel online DRL method enables a UAV to navigate from any starting point to its destination, avoiding obstacles using only raw sensor data. A long short-term memory based subnetwork to solve the sub-problems related to UAV obstacle avoidance in highly dynamic environments is presented in [28]. This subnetwork's actions are evaluated through a comprehensive loss function, and those satisfying the UAV's navigation policy requirements are used to solve the broader obstacle avoidance challenge. A joint MADDPG-based trajectory optimization technique was introduced in [29],

utilizing the collaborative capabilities of MADDPG to guide multiple UAVs. This approach eliminates data isolation through joint averaging, which accelerates convergence. The study [30] proposed a DRL-based UAV coverage aware navigation framework to avoid UAV flying in weak coverage areas of ground BSs. However, it only considers the navigation of a single UAV and ignores noise interference in communication and obstacles in the airspace.

Based on the above-related research, we propose a two-stream graph multi-agent proximal policy optimization (two-stream GMAPPO) algorithm, which integrates MARL with graph neural network (GNN) to address the challenge of multi-UAV cooperative navigation under communication coverage. Graph structures represent the relationships among UAVs, obstacles, and target task points. The GNN processes these graph-structured data by aggregating node information to extract potential features. The extracted features are integrated with the UAV's state space through a two-stream network, enriching local information to enhance environmental awareness, improve decision-making, and ensure safer, more robust navigation. The main contributions of this paper are as follows:

(1) The cooperative navigation problem of multi-UAV systems under communication coverage is formulated as a Markov game. Unlike the study [30], we address the multi-UAV cooperative navigation task in an environment containing static and dynamic obstacles.

(2) The multi-UAV, obstacles and target task points are modeled as UAV-entity graph-structured data, and GNN is employed to extract potential features. This approach enables multi-UAV to obtain richer local information, improving navigation safety and robustness.

(3) We propose a two-stream GMAPPO algorithm for cooperative navigation of multi-UAV systems under communication coverage. The two-stream network structure integrates a GNN module that processes UAV-entity graph-structured data with the algorithm's state space, enhancing state awareness and fostering more effective cooperation compared to the MAPPO [26] algorithm based on the CTDE [24] framework.

The paper is organized as follows: Section II provides the problem description and system modeling. Section III presents the two-stream GMAPPO algorithm proposed in this paper. Section IV analyses the results of the experiment. Section V concludes the paper.

II. PROBLEM DESCRIPTION AND SYSTEM MODELING

In this section, we present the problem of multi-UAV navigation under communication coverage in an urban environment. Then, we describe the communication coverage environment. Finally, we formulate the navigation problem as a Markov game, detailing the composition of the action space, state space, and reward function.

A. Problem Description

In this experiment, we investigate the multi-UAV navigation problem under communication coverage in an urban environment, as shown in Fig. 1. The experimental scenario includes

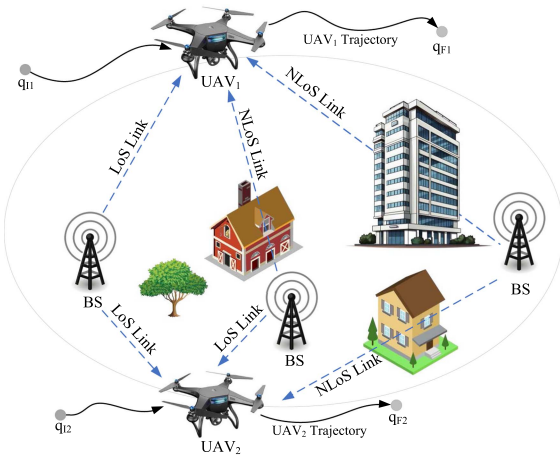


Fig. 1. Multi-UAV navigation system under communication coverage in an urban environment.

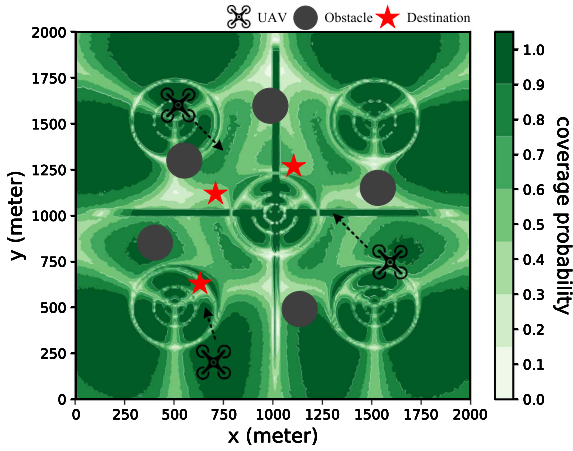


Fig. 2. Communication coverage strength map of ground BSs.

multi-UAV, a set number of obstacles, target task points, and several ground BSs. At the beginning of each episode, the initial positions of the obstacles, target task points, and UAVs are randomly generated, ensuring that there are no overlaps. The positions of obstacles and the target task points are restricted within a region that is a certain distance away from the boundary. Each UAV is randomly assigned a specific target task point in the flight area, and a reward is granted only when it successfully reaches its designated target.

Ground BSs provide communication support for the multi-UAV systems. In this urban environment, five BSs of similar height are deployed, each equipped with three sectors. The communication coverage of the BSs is shown in Fig. 2, where the green shades represent the strength of the coverage, with darker shades indicating stronger signals. At time t , each UAV receives communication support from only one BS sector while flying at a fixed altitude.

The UAV's optimization goal is to maximize its communication coverage strength and minimize the distance to its target task point. Before reaching their target task points, each UAV must ensure optimal communication coverage and maintain a

safe distance from obstacles and other UAVs to avoid collisions. This requires UAVs to have the ability to make autonomous, intelligent decisions in unknown and complex environments.

B. Modeling of the Communication Coverage Environment

In this experiment, each ground BS is positioned at the same height h_{BS} and divided into three sectors, each equipped with a vertically oriented uniform linear array of eight elements. The antenna model is developed in accordance with 3GPP specifications.

The radiation pattern of each element is determined by combining its horizontal and vertical radiation patterns, defined as follows:

$$AH = -\min \left[12 \left(\frac{\phi}{\phi_{3dB}} \right)^2, A_m \right] \quad (1)$$

$$AV = -\min \left[12 \left(\frac{\theta - 90}{\theta_{3dB}} \right)^2, A_m \right] \quad (2)$$

where ϕ represents the azimuth angle, indicating the antenna's angle in the horizontal plane, and θ represents the elevation angle, indicating the antenna's angle in the vertical plane, both measured in degrees. ϕ_{3dB} and θ_{3dB} denote the half-power beamwidths, A_m represents the element gain threshold.

The total gain of the antenna elements is expressed in dB as:

$$\begin{aligned} G_{eledB} &= G_{\max} + A_{ele} \\ &= G_{\max} + \{-\min[-(AH + AV), A_m]\} \end{aligned} \quad (3)$$

where A_{ele} represents the power gain of the antenna element and $G_{\max}$ denotes its maximum directional gain. For simplicity in computation, G_{eledB} is converted to its linear scale G_{ele} .

The combined gain G of the antenna array is expressed in dB as:

$$G = 10 \times \log_{10} |F_{ele} \times AF|^2 \quad (4)$$

where F_{ele} represents the arithmetic square root of G_{ele} , and AF denotes the antenna array factor.

In this experiment, the communication link between the UAV and the BS sector is classified as either a line-of-sight (LoS) or non-line-of-sight (NLoS) link based on whether buildings in the environment obstruct the link, as shown in Fig. 1. The path loss of the LoS link from the UAV to sector k is expressed in dB as:

$$h_k^{\text{LoS}}(t) = 28 + 22 \log_{10} d_k(t) + 20 \log_{10} f_c \quad (5)$$

where $d_k(t)$ represents the distance between the UAV and sector k , and f_c denotes the carrier frequency.

The path loss of the NLoS link between the UAV and sector k is given in dB as:

$$\begin{aligned} h_k^{\text{NLoS}}(t) &= -17.5 + (46 - 7 \log_{10} h(t)) \log_{10} d_k(t) \\ &\quad + 20 \log_{10} (40\pi f_c / 3) \end{aligned} \quad (6)$$

where $h(t)$ is the height of UAV at time t . Additionally, small-scale channel fading $\tilde{h}_k$ is modeled as Rician fading for LoS links and Rayleigh fading for NLoS links.

The baseband equivalent channel between the UAV and the communication BS sector k at time t is denoted by $H_k(t)$, where $1 \leq k \leq K$, and K represents the total number of communication BS sectors linked with the UAV throughout its flight. The baseband equivalent channel $H_k(t)$ is related to the BS antenna gain G_k , the large-scale path loss β_k , and the small-scale fading $\tilde{h}_k$. Both G_k and β_k are functions of the UAV's position $p(t)$ at time t , while $\tilde{h}_k$ is a random variable. The signal power received by the UAV from BS sector k at time t can be expressed as:

$$p_k(t) = \bar{P}_k |H_k(t)|^2 = \bar{P}_k \beta_k(p(t)) G_k(p(t)) \tilde{h}_k(t) \quad (7)$$

where $\bar{P}_k$ represents the transmit power of the BS sector k , and is assumed constant. The path loss is calculated using the following equation:

$$\beta_k(p(t)) = \begin{cases} 10^{-\frac{h_k^{\text{LoS}}(t)}{10}}, & \text{if LoS link} \\ 10^{-\frac{h_k^{\text{NLoS}}(t)}{10}}, & \text{if NLoS link.} \end{cases} \quad (8)$$

In this experiment, the SINR is used as an essential criterion for evaluating the communication coverage performance of UAVs. It can be expressed as:

$$\text{SINR}(t) \triangleq \frac{p_{I(t)}(t)}{\sum_{m \neq I(t)} p_m(t) + \sigma^2} \quad (9)$$

where $I(t)$ denotes the sector that provides service for the UAV at time t , and m represents the BSs not associated with the UAV at that time. In this case, the communication of the UAV is affected not only by interference from all non-associated BS sectors but also by the environmental noise, which impacts the quality of its communication.

Each UAV has an independent SINR at time t . To maintain communication coverage while in flight, the UAV's SINR must remain above a minimum threshold α . If the SINR at time t falls below α , the UAV is considered to be outside the communication coverage of the BS.

C. Markov Decision Process and Markov Game

Markov decision process (MDP) is a tuple defined by five data of $\langle S, A, R, P, \gamma \rangle$, which includes a state space S , an action space A , a reward function R , a state transfer matrix function P , and a discount factor γ . At time t , during the agent's interaction with the environment, the agent first observes the environment and receives its current state. Then, it selects an action to execute from the action space according to the policy π learned at this time. At this point, the environment shifts to a new state, determined by the probability distribution $P(s|s, a)$, and the agent receives a new reward $R(s, a)$.

The Markov game [31] represents a significant extension of the MDP framework for multi-agent systems. Multiple agents interact within a shared environment, each aiming to maximize their expected cumulative discounted reward. The state space S , action space A , and reward function R are expanded to accommodate the presence of multiple agents.

For a random policy, the state value function, which represents the expectation of the cumulative discounted reward obtained by an agent picking an action and executing by the policy π , starting

from state s_t , is expressed as follows:

$$V_\pi(s_t) = \mathbb{E} \left[\sum_{l=0}^{\infty} \gamma^l r(s_{t+l}, a_{t+l}) \mid s_t \right]. \quad (10)$$

Similarly, its action value function, representing the expectation of the cumulative discounted rewards obtained by the agent acting by the policy π after the agent selects and executes the action a_t starting from the state s_t , has the following expression:

$$Q_\pi(s_t, a_t) = \mathbb{E} \left[\sum_{l=0}^{\infty} \gamma^l r(s_{t+l}, a_{t+l}) \mid s_t, a_t \right]. \quad (11)$$

The objective for all agents is to learn an optimal policy π^* that maximizes the total expected reward. This optimal policy is denoted as:

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]. \quad (12)$$

D. The State Space and Action Space for Multi-UAV

In DRL, the UAV's decision-making process relies heavily on the state information it perceives from the environment. Therefore, designing an appropriate state space is essential for enabling UAVs to make informed decisions and select optimal actions. The UAV's state space should accurately represent its status as well as crucial information about the environment, such as obstacles and target task points.

The state space is defined as $S = (s_1, s_2, \dots, s_N)$, where the state of each UAV is given by $s_i = (s_{Pi}, s_{Ei}, s_{Gi})$. Specifically, $s_{Pi} = (x_i, y_i, v_{xi}, v_{yi}, \text{SINR}_i)$ describes the UAV's position (x_u, y_u) , velocity (v_{xu}, v_{yu}) , and SINR. The component $s_{Ei} = (l_{gi}, l_{oi})$ denotes the UAV's observed distances to the target task point and nearby obstacles. $s_{Gi} = (\text{node}, \text{adj})$ represents the observation associated with the graph, where node denotes the observation of each node and adj denotes the adjacency matrix of the graph.

The action space $A = (a_1, a_2, \dots, a_N)$ consists of the individual actions available to each UAV within a multi-agent system. For this experiment, the UAV's motion is simplified. Each UAV has a discrete action space, with its initial velocity defined as $a_i = (v_x, v_y)$. The UAV's velocity changes only due to external factors, such as collisions. If a UAV collides, its velocity may change due to the impact force and its mass.

E. Reward Function

The reward function is a crucial component of DRL, significantly influencing algorithm performance and guiding the agent in learning an effective behavioral policy. The reward function is the most crucial part of DRL. It is typically represented as $R(s, a)$, where s represents the current state and a is the action taken, the reward function provides feedback from the environment based on the action performed. A high reward indicates that the decision is advantageous, increasing the likelihood that the agent will choose similar actions in future policy updates.

The UAV's reward function in this experiment consists of four main components. First, the reward depends on the SINR

achieved by the UAV at time t , represented by R_{SINR} , which encourages the UAV to fly in regions with better communication coverage. Second, the reward includes the negative Euclidean distance between the UAV and the target task point, promoting faster arrival at the destination. Third, to ensure safe flight, a constant penalty r_{coll} is applied whenever the distance between the UAV and an obstacle falls below the safety threshold. Finally, the UAV incurs a penalty r_{out} if it leaves the designated task area. The reward function is defined as:

$$r(s_t, a_t) = R_{SINR} + R_{dist} + r_{coll} + r_{out}. \quad (13)$$

The R_{SINR} is calculated as follows:

$$R_{SINR} = \kappa \cdot SINR(t) \quad (14)$$

where κ is a constant coefficient about $SINR(t)$. The R_{dist} between the UAV and the target task point is computed as:

$$R_{dist} = \begin{cases} -\sqrt{(x_u - x_{tar})^2 + (y_u - y_{tar})^2}, & \text{if } l \geq d \\ r_{goal}, & \text{if } l < d \end{cases} \quad (15)$$

where (x_u, y_u) is the coordinate information of the UAV, (x_{tar}, y_{tar}) is the coordinate information of the target task point, and l is the distance from the UAV to the target task point. As the UAV approaches the target task point, it will receive a greater reward. When the distance is less than the threshold d , it is determined that the UAV has reached the target point, giving a reward r_{goal} .

III. MULTI-UAV COOPERATIVE NAVIGATION UNDER COMMUNICATION COVERAGE ALGORITHM BASED ON GRAPH NEURAL NETWORK

In this section, we first introduce the PPO algorithm and the MAPPO algorithm integrated with the CTDE framework. Then, we present the GNN. Finally, we provide a detailed explanation of the two-stream GMAPPO algorithm proposed in this paper.

A. Proximal Policy Optimization Algorithm

The PPO [23] algorithm has its origins in the PG [20] algorithm, which is essentially based on the actor-critic framework. The PPO algorithm introduces proximal optimization to enhance sample utilization efficiency. This is achieved by regulating the magnitude of updates to prevent policy updates from being excessively drastic. Simultaneously, gradient clipping is employed to smooth strategy updates, reducing variance in training and thereby enhancing the overall stability of the algorithm.

The PPO algorithm employs direct truncation on the ratio of old and new policies to limit the magnitude of change. Its objective function is computed as follows:

$$L^{Clip}(\theta) = \mathbb{E}_t \left[[\min(r_t(\theta) \hat{A}_t), \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)] \right] \quad (16)$$

where $\hat{A}_t$ represents the advantage function; $r_t(\theta)$ represents the ratio of old to new policies; ϵ represents the truncation hyperparameter, which is a constant; $\text{clip}(\cdot)$ denotes the truncation function, which is responsible for restricting the ratio $r_t(\theta)$ to be within the interval $[1 - \epsilon, 1 + \epsilon]$ in order to ensure convergence;

and ultimately, $L^{Clip}(\cdot)$ forms a lower bound on the objective using the min function that selects a smaller value between the untruncated and truncated objective.

If the advantage function $\hat{A}_t$ is positive, it is essential to increase the old-to-new policy ratio $r_t(\theta)$. However, when $r_t(\theta) > 1 + \epsilon$, no additional incentive will be provided. Conversely, if the advantage function $\hat{A}_t$ is negative, it is necessary to decrease the old-to-new policy ratio $r_t(\theta)$. However, when $r_t(\theta) < 1 - \epsilon$, no additional incentive will be provided. This approach ensures that the difference between the old and new policy is limited to a reasonable extent.

The critic network is updated using the gradient descent method with the following loss function:

$$L^{vf}(\phi) = \mathbb{E}_t [(V_\theta(s) - V_{target})^2] \quad (17)$$

where $V_\theta(s)$ is the value function of state S and V_{target} is the target value function.

The total loss function of the PPO algorithm can be defined as:

$$L = \mathbb{E}_t [L^{Clip}(\theta) - c_1 L^{vf}(\phi) + c_2 S[\pi_\theta]] \quad (18)$$

where c_1 and c_2 are the hyperparameters used to balance the weights of the different loss functions, and $S[\pi_\theta]$ is the entropy of the policy, which is used to increase the employability of the policy.

B. Multi-agent Proximal Policy Optimization Algorithm

When applying the PPO [23] algorithm directly to multi-agent systems, it faces the challenge of non-smooth and uncoordinated agent behavior. To address this challenge, the MAPPO [26] algorithm is introduced, which integrates the CTDE [24] framework into the PPO algorithm.

In the MAPPO algorithm, each agent is equipped with a local critic network capable of observing global information, including information from other agents. This network receives observations and actions from all the agents and learns them as joint state-action pairs. At this point, the critic network learns a centralized value function rather than the value function of the agents themselves. As for the actor network, each agent formulates its policy only through its local observation state. This design aims to enable the agents to learn in the direction of optimizing the global value function.

The actor network optimization objective is:

$$L(\theta) = \left[\frac{1}{Bn} \sum_{i=1}^B \sum_{k=1}^n \min(r_{\theta,i}^k, A_i^k), \text{clip}(r_{\theta,i}^k, 1 - \epsilon, 1 + \epsilon) A_i^k \right] + \sigma \frac{1}{Bn} \sum_{i=1}^B \sum_{k=1}^n S[\pi_\theta(o_i^k)] \quad (19)$$

where $r_{\theta,i}^k$ denotes the weight of the old and new policies; A_i^k represents the advantage function; S denotes the entropy of the policy; σ is a hyperparameter controlling the entropy coefficient.

The critic network optimization objective is:

$$L(\phi) = \frac{1}{Bn} \sum_{i=1}^B \sum_{k=1}^n \max([V_\phi(s_i^k) - \hat{R}_i]^2, \\ (\text{clip}(V_\phi(s_i^k), V_{\phi_{old}}(s_i^k) - \varepsilon, V_{\phi_{old}}(s_i^k) + \varepsilon) - \hat{R}_i)^2] \quad (20)$$

where $\hat{R}_i$ is the discount reward.

C. Graph Neural Network

GNN is a method for modeling and processing graph-structured data. Unlike traditional neural networks, GNN efficiently capture relationships between nodes and edges in a graph, enhancing the inference and prediction of graph structures [32]. These models demonstrate strong relational induction capabilities when processing graph data, allowing them to learn and comprehend complex relationships between nodes and establish rules and patterns for the graph's structure.

The fundamental principle of the GNN model is as follows: for a given graph G comprising m nodes (denoted as $G = (A, X)$), where $A \in \{0, 1\}^{(m \times m)}$ is the adjacency matrix representing node-to-node connections, and $X \in \mathbb{R}^{(m \times d)}$ denotes the node features, with d being the dimension of the feature vector for each node. In GNN, the process of aggregating information can be described as:

$$X_{i+1} = \sigma(D^{(-\frac{1}{2})} \hat{A} D^{(-\frac{1}{2})} X_i W_i) \quad (21)$$

where $\hat{A}$ denotes introducing a self-loop mechanism based on adjacency matrix A to ensure each node considers its information; D represents the diagonal matrix of node degrees, calculated after normalizing the adjacency matrix A . The node features are scaled by taking the negative square root of A to ensure numerical stability; X_i denotes the embedding vectors of the nodes, representing the learned node features; W_i is the learnable weighting matrix used to adjust the linear combination of node features; σ is a nonlinear activation function, which introduces a nonlinear transformation to enhance the expressiveness of the network.

GNN comprises three primary components: (1) Initialization: Begin by assigning each node's feature vector using the node attributes. (2) Neighborhood Detection: Identify the immediate neighbors of each node to gather supplementary data. (3) Information Aggregation: Enhance a node's feature vector by consolidating and condensing the features of its identified neighbors.

D. Two-stream Graph Multi-agent Proximal Policy Optimization Algorithm

A crucial aspect of addressing multi-UAV cooperative navigation tasks is understanding the interaction relationships between UAVs and environment. However, the dynamic nature of the environment and the constant movement of UAVs make it challenging to represent these interactions. This experiment exploits the inherent structure of graph-structured data to capture UAV interaction relationships by constructing an UAV-entity graph.

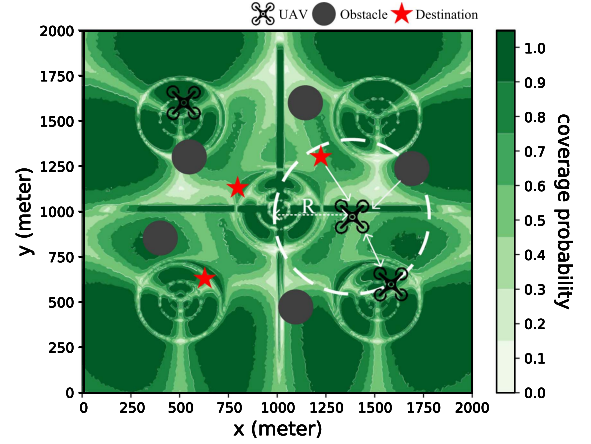


Fig. 3. UAV-entity graph under communication coverage environment.

In this experiment, each object in the communication coverage navigation environment, including UAVs, target task points, and obstacles, is defined as an entity. At each time t , a UAV-entity graph-structured data about UAV i , denoted as $G_i = (V, E)$, where each node V represents an entity in the environment. We assume that UAV can only observe the presence of obstacles or other UAVs within a specific finite radius R , as shown in Fig. 3. If the distance between the UAV and an entity is within this range, an edge E exists connects them. If the entity is a UAV, the edge is bidirectional, allowing messages to be exchanged between them. If the entity is an obstacle or a target task point, this edge is unidirectional, permitting messages to flow only from the entity to the UAV.

The features of each node j in graph-structured data G_i are defined as:

$$x_j = [p_i^j, v_i^j, p_i^{goal}, type] \quad (22)$$

where p_i^j , v_i^j , and $type$ represent the relative position of UAV i with its associated entity of node j , its relative velocity, and the category of the entity. If the entity is a UAV, the p_i^{goal} indicates the distance from the UAV to its target task point; otherwise the p_i^{goal} remains the relative position of the entity to the UAV.

Each edge E in the graph-structured data G_i is assigned an edge feature, which is represented by the adjacency matrix. Instead of the traditional approach, which uses only 0 and 1 to indicate the presence of an edge between two nodes, we assign the modal value of the relative distance between nodes as the edge weight. This method ensures that the model better captures the spatial relationships between nodes in the graph. Using these edge weights provides richer information to the GNN, enhancing its ability to consider distance relationships between nodes in this experiment.

In this experiment, the UAV node and edge features are transformed and updated through the embedding layer. This process preserves the network's topology and node information, while effectively capturing the underlying structure and relationships between nodes.

After this transformation, information aggregation takes place using the unified message passing (UniMP) model within the

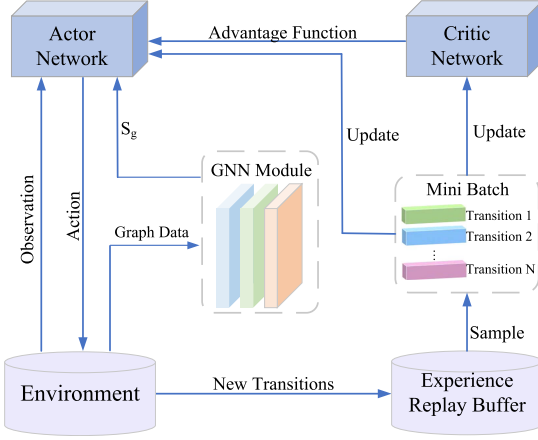


Fig. 4. Framework of two-stream GMAPPO algorithm.

GNN. UniMP is a variant of the graph transformer, incorporates an attention mechanism that allows nodes to prioritize messages from neighboring nodes based on their importance. This enables UAVs to focus on local information, improving situational awareness, reducing collision risks, and enhancing flight safety. We use a two-layer GNN architecture, as deeper networks can degrade performance by overfitting or causing feature divergence. The update of each layer of UniMP is defined as:

$$x'_i = W_1 x_i + \sum_{j \in N(i)} \alpha_{i,j} W_2 x_j \quad (23)$$

where x_i represents the node features in the graph; $N(i)$ represents the set of nodes connected to node i ; W_k denotes the learnable weight matrix. $\alpha_{i,j}$ are the attention coefficients computed by multi-head dot product attention:

$$\alpha_{i,j} = \text{softmax} \left(\frac{(W_3 x_i)^T (W_4 x_j + W_5 e_{ij})}{\sqrt{c}} \right) \quad (24)$$

where e_{ij} represents the edge features connecting nodes i and j , and c denotes the dimension of the output layer. Through the processing of the UniMP module for deep feature extraction of the graph-structured data, UAV i aggregates the observation information from other UAVs and obtains its own node feature vector S_g .

As shown in Fig. 4, the two-stream network structure processes the aggregated node feature S_g from the preceding GNN, which captures the interactions between UAVs and other entities. These features are passed to the next network for further decision-making. In the actor network, the UAV's local observation is concatenated with the node feature S_g to form the input. This approach offers greater flexibility than traditional inputs, adapting more effectively to UAV position changes and responding better to the movement patterns of other UAVs. As a result, the actor network can make more informed decisions, generate safer maneuvers, and proactively avoid potential collisions. The node feature S_g is combined with global observation information in the critic network. This design integrates both node feature S_g and global observation, allowing the network to evaluate

Algorithm 1: Two-Stream GMAPPO Algorithm.

```

Initiate policy network  $\pi_\theta$  and value network  $V_\omega$ , let
 $\pi_{\theta'} = \pi_\theta$ .
Initiate a memory buffer  $D$ .
for episode  $l = 1, 2, \dots, L$  do
   $s_{o,1}$  = initiate state
   $s_{g,1} = \text{GNN}(G = (V, E))$ 
  for step  $t = 1, 2, \dots, T$  do
     $s_t = s_{o,t} + s_{g,t}$ 
    Each UAV executes action according to  $\pi_{\theta'}(a_t | s_t)$ 
    Get the reward  $r_t$ , and the next observation  $s_{o,t+1}$ ,
    and the node feature  $s_{g,t+1}$ 
  end for
  Get a trajectory for each UAV:  $\tau = \{s_t, a_t, r_t, s_{g,t}, s_{o,t}\}_t^T$ 
  Compute  $\{\hat{Q}(s_t, a_t)\}_{t=1}^T$ 
  Compute advantages  $\{A_t(s_t, a_t)\}_{t=1}^T$ 
  Store data  $\{[s_t, a_t, \hat{Q}(s_t, a_t), A_t(s_t, a_t)]^N, s_{g,t}, s_{o,t}\}_t^T$ 
  for  $k = 0, 1, 2, \dots, K$  do
    Shuffle and renumber the data's order
    for  $j = 0, 1, 2, \dots, \frac{T}{B} - 1$  do
      Select  $B$  group of data  $D_j$ :
       $D_j = \{s_t, a_t, \hat{Q}(s_t, a_t), A_t(s_t, a_t), s_{g,t}, s_{o,t}\}^{B(q+1)}$ 
      for  $j = 0, 1, 2, \dots, M$  do
         $\Delta \theta^j = \frac{1}{B} \sum_1^B \{\nabla_{\theta^j} f(r(\theta^j), A^j(s, a))\}$ 
         $\Delta \omega^j = \frac{1}{B} \sum_1^B \{\nabla_{\omega^j} (\hat{Q}^j(s_t, a_t) - Q_{\omega^j}(s, a))^2\}$ 
        Apply gradient ascent on  $\theta^j$  and  $\Delta \theta^j$  by Adam
        Apply gradient descent on  $\omega^j$  and  $\Delta \omega^j$  by Adam
      end for
    end for
  end for
  for  $j = 0, 1, 2, \dots, M$  do
    Update  $\theta_{old}^j \leftarrow \theta^j$  and  $\omega \leftarrow \omega^j$ 
  end for
  Empty  $D$ 
end for

```

the value of multi-UAV actions more accurately. Combining the node feature S_g enhances the critic network's ability to evaluate the effectiveness and accuracy of UAV actions.

IV. SIMULATION RESULTS

In this section, we describe the parameter settings in the simulation environment and conduct the corresponding simulation experiments. By comparing with the MARL baseline algorithms, we successfully demonstrate the effectiveness of the proposed two-stream GMAPPO algorithm.

A. Experiment Setting

In this experiment, we construct a 2 km $\times$ 2 km urban area scenario containing numerous buildings. The distribution of these buildings is primarily determined by three parameters: α_{bd} , β_{bd} , and γ_{bd} , where $\alpha_{bd} = 0.3$, $\beta_{bd} = 300$, and $\gamma_{bd} = 50$. The parameter α_{bd} represents the proportion of the total land area

TABLE I
ALGORITHM PARAMETERS SETTING

Definition	Value	Definition	Value
Max episode	60000	Max step per episode	25
Replay buffer capacity	1000000	Minibatch size	32
PPO_epoch	15	GAE parameter	0.95
GNN hidden size	16	MLP hidden size	64

occupied by buildings, and β_{bd} indicates the average number of buildings per square kilometre, respectively. The parameter γ_{bd} follows a Rayleigh distribution to determine building heights, which are limited to a maximum of $h_{bd} = 90$ m.

The presence of a LoS link is determined by assessing whether the communication path between the UAV and the ground BS remains unobstructed, considering the building distribution. The BS in the centre of Fig. 2 is located on a crossroads road in an urban environment, which produces a cross-shaped area of strong communication coverage due to the absence of building obstructions. The scenario includes five ground BSs, resulting in a total of $K = 15$ sectors. Each sector operates with a transmission power of $\bar{P} = 20$ dBm. The half-power beamwidths in both ϕ_{3dB} and θ_{3dB} are set to 65° . The element gain threshold A_m is 30 dB. The maximum directional gain $G_{\max}$ is 8 dBi. The carrier frequency f_c is 8 GHz. The SINR interruption threshold is defined as $\gamma_{th} = 0$ dB, and the noise power is $\sigma^2 = 5$ dBm.

This experiment involves three UAVs, three target task points, three static obstacles, and three dynamic obstacles. Table I presents the parameters for the two-stream GMAPPO algorithm and the MARL baseline algorithms.

B. Impact of Hyperparameters on Two-stream GMAPPO

In this section, we evaluate how different hyperparameters affect the convergence speed and efficiency of the two-stream GMAPPO algorithm, with a particular focus on the reward value of each episode. The hyperparameters analyzed include learning rate, discount factor. To ensure the reliability of the comparison, all other hyperparameters are kept constant, and only one hyperparameter is changed at a time. This approach allows us to isolate and understand the specific impact of each hyperparameter.

Fig. 5 presents the mean reward trends of the two-stream GMAPPO algorithm under different learning rates lr . The learning rate of 0.001 boosts early stage performance gains but causes instability, marked by significant fluctuations in the reward curves. When the learning rate is set to 0.0001, the algorithm encounters difficulties in convergence, resulting in slower and less stable progress. The learning rate of 0.0005 offers an optimal trade-off between efficiency and stability, yielding the highest mean reward. These results emphasize the importance of selecting an appropriate learning rate for MARL.

Fig. 6 shows the influence of discount factors γ on the algorithm's mean reward. A higher discount factor accelerates early convergence and prioritize long-term rewards, making the final converged reward value larger. In contrast, a lower discount

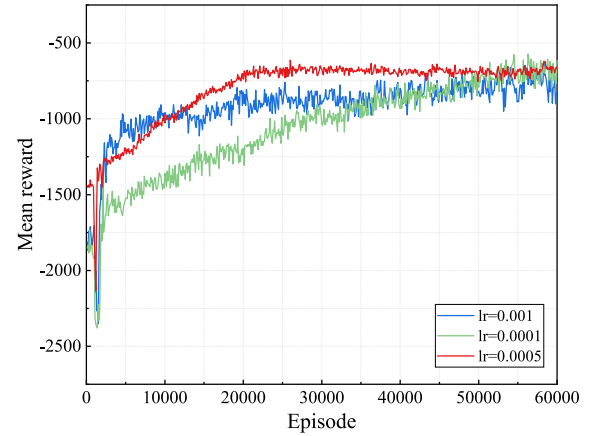


Fig. 5. Mean reward of different learning rates for two-stream GMAPPO.

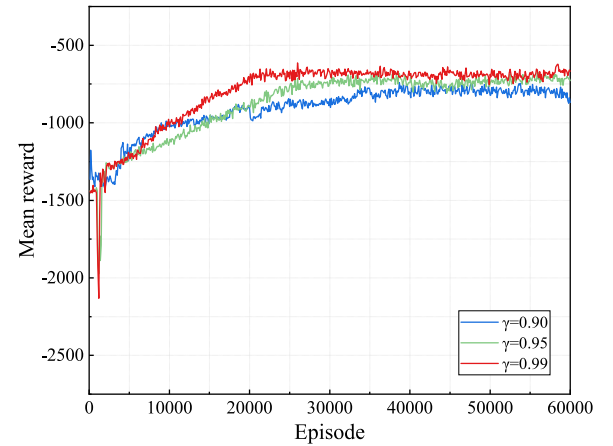


Fig. 6. Mean reward of different discount factors for two-stream GMAPPO.

factor focuses on short-term results, leading to a lower final reward.

Fig. 5 and Fig. 6 highlight the critical role of learning rate and discount factor selection in multi-UAV navigation tasks. The carefully optimized combination of these hyperparameters balances stability with convergence speed. Based on the above experiments, the learning rate of the two-stream GMAPPO algorithm in this study is set to 0.0005 and the discount factor is set to 0.99.

C. Result Analysis of Different MARL Algorithms

To validate the effectiveness of the proposed two-stream GMAPPO algorithm, we compared it with three MARL baseline algorithms: MADDPG [24], MATD3 [25], and MAPPO [26]. All algorithms were tested with identical hyperparameter values to ensure fair comparison.

Fig. 7 shows the mean reward trends of MARL algorithms over 60000 training episodes in a multi-UAV cooperative navigation task. The mean reward per episode reflects the algorithms' convergence and learning performance. Higher mean rewards signify improved performance in communication coverage and

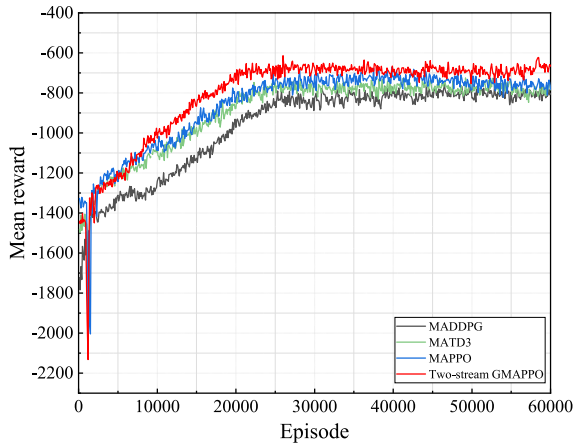


Fig. 7. Mean reward for different MARL algorithms.

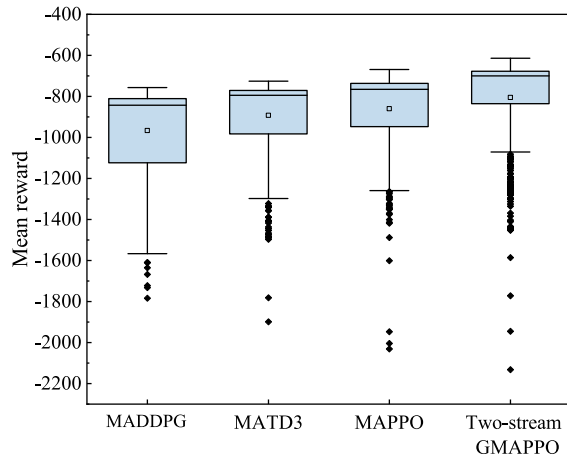


Fig. 8. Mean reward distribution for different MARL algorithms.

obstacle avoidance. Early in training, all algorithms exhibit a decline in rewards as they prioritize exploration. This exploration is crucial for UAVs to build situational awareness and improve long-term performance. The two-stream GMAPPO algorithm converges around 24,000 episodes, stabilizing at a mean reward of approximately -700, outperforming the other algorithms. This result suggests that two-stream GMAPPO adapts faster to complex communication environments and achieves superior navigation paths. In contrast, other algorithms converge more slowly and at lower reward values, exposing the limitations of traditional MARL approaches for this task. The MADDPG algorithm performs the worst, displaying unstable training with reward values around -880.

For a clearer performance comparison, Fig. 8 provides box-and-line plots of the mean rewards per episode for each algorithm. This figure illustrates the mean reward distribution for MARL algorithms in the multi-UAV cooperative navigation task. The two-stream GMAPPO algorithm shows the most compact reward distribution and the highest median, demonstrating consistent performance with minimal fluctuations across episodes. Additionally, two-stream GMAPPO has very few outliers, demonstrating its robustness in complex environments.

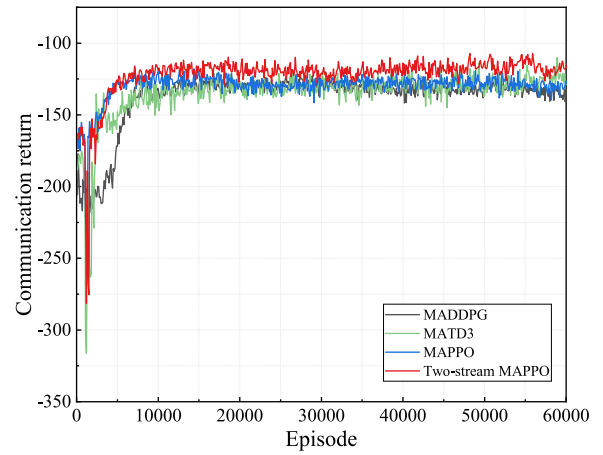


Fig. 9. Communication return for different MARL algorithms.

The MATD3 and MADDPG algorithms exhibit more scattered reward distributions with a greater number of low values, indicating their lack of stability in the navigation task. Although the median rewards of MAPPO are similar to those of two-stream GMAPPO, its wider distribution indicates more instability compared to two-stream GMAPPO.

The superior performance of two-stream GMAPPO is due to its integration of GNN, which captures complex relationships among UAVs, obstacles, and target task points. The GNN module improves the algorithm's ability to aggregate neighborhood information, enabling UAVs to detect environmental changes more accurately and reduce collision risks. Additionally, the two-stream network structure avoids local optima during training, enhancing stability. These improvements allow two-stream GMAPPO to surpass other algorithms in convergence speed and final mean reward, highlighting its potential for collaborative multi-UAV systems.

Fig. 9 illustrates the variation in communication return across training episodes for a multi-UAV system using different MARL algorithms. The communication return evaluates each algorithm's ability to maintain UAVs within areas of strong communication coverage in complex environments. In the early training process, communication returns are low and fluctuating because the algorithms are in the exploratory phase, executing many sub-optimal policies. Among the algorithms, two-stream GMAPPO performs best, with communication return converging at around 8,000 episodes and stabilizing near -120. In comparison, other algorithms converge more slowly and with lower final values. MADDPG and MATD3 not only achieve lower communication returns than two-stream GMAPPO and MAPPO but also display significant fluctuations, reflecting reduced adaptability to complex environments. The superior performance of two-stream GMAPPO and MAPPO is attributed to their policy clip operation, which reduces abrupt policy changes and improves stability.

Fig. 10 presents the variation in collision numbers across training episodes for different MARL algorithms in a multi-UAV navigation task. The number of collisions is a key safety metric in complex environments. Early training sees frequent collisions as

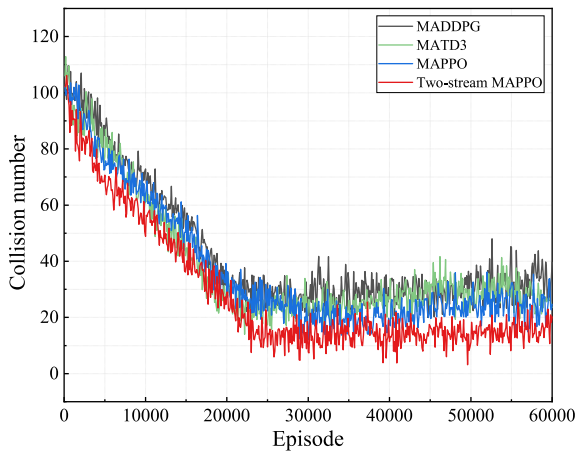


Fig. 10. Collision number for different MARL algorithms.

algorithms explore, but two-stream GMAPPO reduces collisions significantly and stabilizes quickly. The MAPPO algorithm also reduces collisions but experiences instability in certain episodes. The MADDPG and MATD3 show consistently higher collision rates, indicating poor performance in dynamic interactions and obstacle avoidance.

The advantage of two-stream GMAPPO stems from the collaboration between its two-stream network structure and the GNN module. This integration effectively models and aggregates information from UAVs and other entities, allowing UAVs to perceive the environment precisely and adjust flight strategies to avoid collisions and optimize flight paths. In comparison, other algorithms struggle with the efficient transmission and processing of environmental information, limiting their ability to enhance communication returns and mitigate collisions in complex environments.

V. CONCLUSION

In this study, we have introduced a collaborative navigation problem for multi-UAV systems under communication coverage in an urban environment. We have formalized the navigation problem as a Markov game and propose a two-stream GMAPPO algorithm. This algorithm employs a two-stream network structure, modeling entities in the environment as graph-structured data and utilizing GNN for information aggregation. This approach enhances the UAVs' ability to perceive environmental changes and reduces the likelihood of collisions. We have validated the effectiveness of the proposed approach through simulation experiments, comparing it with other baseline algorithms. Experimental results have demonstrated the superiority of the two-stream GMAPPO algorithm, which significantly outperforms other algorithms in convergence speed and mean reward. It also excelled in terms of communication return and reduces the occurrence of collisions.

In future research, we plan to explore more complex tasks such as navigation involving dynamic task points. This will allow us to better understand the robustness and adaptability of our algorithms in scenarios with changing obstacles and conditions.

In addition, we will consider how to improve communication in more challenging scenarios.

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